



Items and Inventory **Units of Measure Setup**

SAP Business One
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PUBLIC

THE BEST RUN 

Welcome to the topic on setting up units of measure and units of measure groups.

Objectives



At the end of this topic, you will be able to:

- Set up unit of measure groups

In this topic, you will learn how to set up unit of measure groups.

Business Scenario



- OEC Computers buys cables for resale. The cables are bought, stored and sold in different units of measure.
- They would like to set up a unit of measure group that contains all the units of measure used for cables.
- They would like to have the unit of measure group with its list of units to be automatically available when each new cable item is created.
- They would also like to set defaults for most frequently used units in sales and purchasing.

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Business Example – Cable Items

OEC Computers sells 3 types of cables.
The items are stored in inventory by meters.
They are purchased in reels.
Each item can be sold three ways: by the meter, in reels and in spools.
We will set up the units of measure then group them together to assign them to the items.

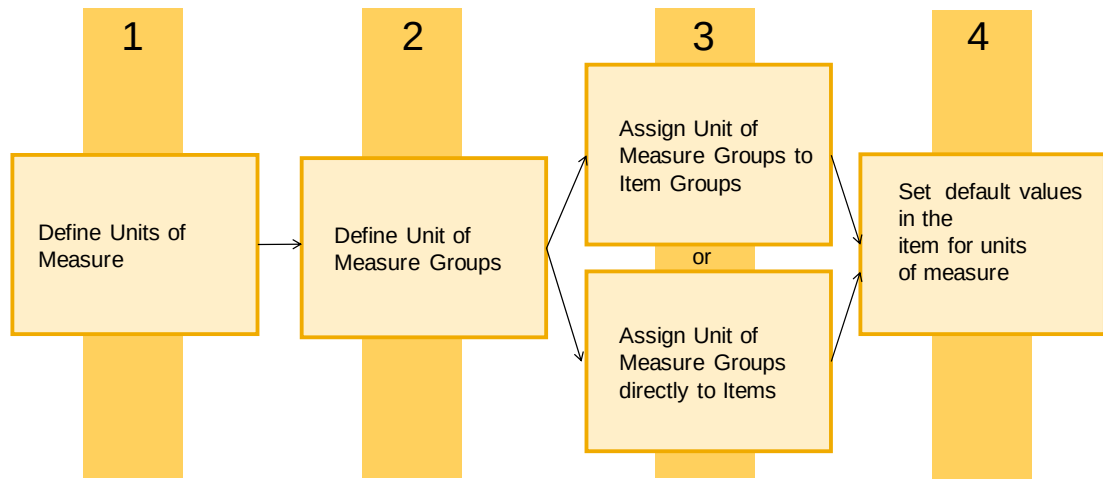


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- OEC computers resells 3 different types of cables.
- All three cable items are stored in inventory by meters.
- The cables are purchased in reels of 100 meters.
- Each cable item can be sold three ways: by any number of meters (cut to measure), in 100 meter reels or in 50 meter spools.
- In our business example we will set up the units of measure that are used by our cable items, then we will group them together into a Unit of Measure Group that we can assign to the items.
- Then each cable item will contain a full list of all units of measure that these items use in transactions. Additionally we will be able to set up relationships from each of these units to barcodes, prices and packaging needed for the items.

Steps to Set up UoMs for Items



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- There are four main steps for setting up units of measures (often abbreviated to UoM) for items.
- First, define the list of all possible units of measure. This is a global list for your company. If you want to use a unit in a document, it must be included in this list.
- Second, decide on how you wish to combine the units into unit of measure groups. The unit of measure groups should be created with some thought as to how items are packaged when they are purchased, sold or managed in inventory. You want to define a unit of measure group that can be used for all items of a similar type.
- Once you have created the groups you can assign them to an item group or assign them directly to items. The advantage of assigning the unit of measure group to an item group is that the unit of measure group will be assigned to an item automatically whenever it is created within that item group.
- Finally, once a unit of measure group is assigned to an item, you can set default values for units of measure for each item. You can set default values for units in purchasing, sales, and inventory as well as in packaging.

Step 1: Define Units of Measure

Administration → Setup → Inventory → Units of Measure

#	UoM Code	UoM Name	Length	Width	Height	Volume	Volume UoM	Weight
1	Manual	Manual					cm	
2	Pack	Pack	30cm	21cm	4cm	2,520	cc	2.5kg
3	6Pack	6 Pack	30cm	21cm	24cm	15,120	cc	15kg
4	Carton	Carton	60cm	42cm	24cm	60,480	cc	60kg
5	Pallet	Pallet	84cm	60cm	24cm	120,960	cc	120kg
6	SmPack	Small Pack	30cm	21cm	2cm	1,260	cc	1.25kg
7	Case	Case					ci	
8	DVD	DVD					ci	
9	Kit	Kit					ci	
10	Each	Eac					cm	
11	Meter	Melter	100cm				cm	
12	Reel	Reel					cm	
13	Spool	Spool					cm	
14							cm	

All units of measure used within the company are defined here.

The first step for using units of measure is to define a global list of units of measure. All units of measure used within the company are defined here.

To define the units of measure open the Inventory Setup menu in the Administration module and choose Units of Measure.

In this window, we define a general list of all the units of measures in the system.

In our business example, we would like to set up unit of measure codes and groups for the cables we buy and sell.

Since we store and sell our cables by length in meter units, we enter a unit code and unit name for meter.

This new unit will be available for us to choose in the Item Master Data once we have added it to the unit of measure group of the item that we assign to the item.

We will also need to add any other units we plan to use in transactions for cables, such as reels and spools.

Add Dimensions to Units

Administration → Setup → Inventory → Units of Measure

#	UoM Code	UoM Name	Length	Width	Height	Volume	Volume UoM	Weight
1	Manual	Manual					cm	
2	Pack	Pack	30cm	21cm	4cm	2,520	cc	▼ 2.5kg
3	6Pack	6 Pack	30cm	21cm	24cm	15,120	cc	▼ 15kg
4	Carton	Carton	60cm	42cm	24cm	60,480	cc	▼ 60kg
5	Pallet	Pallet	84cm	60cm	24cm	120,960	cc	▼ 120kg
6	SmiPack	Small Pack	30cm	21cm	2cm	1,260	cc	▼ 1.25kg
7	Case	Case					cl	▼
8	DVD	DVD					cl	▼
9	Kit	Kit					cl	▼
10	Each	Eac					cm	▼
11	Meltier	Meltier	100cm				cm	▼
12	Reel	Reel					cm	▼
13	Spool	Spool					cm	▼
14							cm	▼

Optional:

Add dimensions for each unit of measure.

- In the same window, we can also enter optionally dimensions for each unit of measure. Dimensions are very useful for determining packaging for items.
- Dimensions can be entered using characteristics such as length, width, height, and weight. Volume is automatically calculated based on the length, width and height.
- Here we see that some of the codes have these dimensions specified while other do not.
- It is not mandatory to enter values for the dimensions, but it can be helpful. The dimensions entered here will be the default values for these units when they are chosen in the item master data. The dimensions can be changed in the item master.
- However if you frequently use the same unit of measure for many different items with different dimensions, it is recommended to leave these fields blank.

Characteristics for Global Units

Administration → Setup → Inventory → Length and Width / Weight

The image shows two side-by-side screenshots of SAP configuration windows. The left window is titled 'Length and Width - Setup' and contains a table with two columns: '# Code' and 'Unit Name'. It lists units from 1 to 6: 1. Inch, 2. Foot, 3. Centimetre, 4. Decimetre, 5. Metre, and 6. Millimetre. The right window is titled 'Weight - Setup' and contains a similar table with columns '# Code' and 'Unit Name', listing units from 1 to 6: 1. Pound, 2. Ounce, 3. Gramme, 4. Kilogramme, 5. Milligramme, and 6. (empty). Both windows have 'OK' and 'Cancel' buttons at the bottom.

#	Code	Unit Name
1	"	Inch
2	'	Foot
3	cm	Centimetre
4	dm	Decimetre
5	m	Metre
6	mm	Millimetre
7		

#	Code	Unit Name
1	Lb	Pound
2	Oz	Ounce
3	g	Gramme
4	kg	Kilogramme
5	mg	Milligramme
6		

In some cases, you may want to view or update the characteristics for system dimensions before adding new global units of measure. Under the Inventory Setup menu, you will find two windows where you can enter characteristics for the dimensions. One window is to define characteristics used for length and width and the other is for characteristics for measuring weight.

Unit of Measure Group Concept

Global Units of Measure

Pack
Small Pack
6-pack
Pallet
Carton
Each
Case
Meter
Reel
Spool
...

List of UoM Groups

Paper

Media

...

Paper UoM Group

Pack = 21x30x4 cm
Small Pack = 0.5 Pack
6-Pack = 6 Pack
Carton = 24 Pack
Pallet = 48 Pack

Media UoM Group

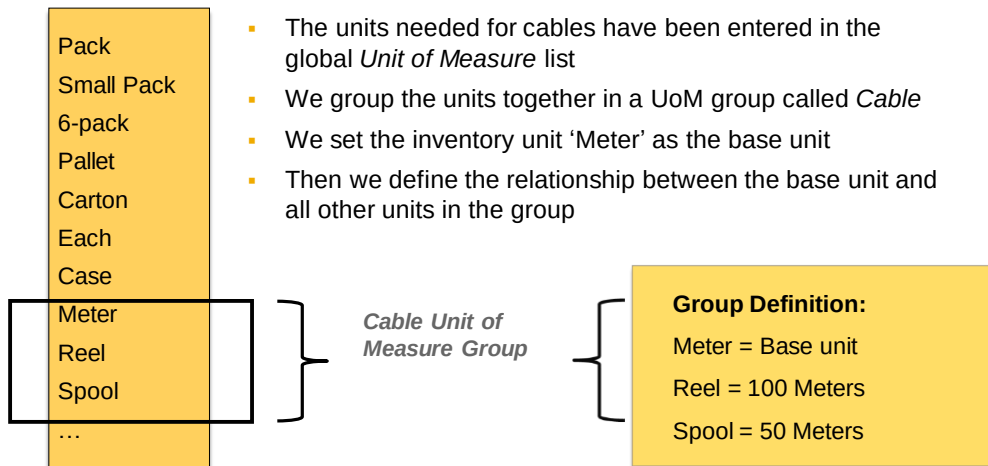
Each = Base unit
Case = 100 X Each
Carton = 10,000 X Each

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- Before we move to step 2 where we set up unit of measure groups. Let's take a moment to better understand the relationship between units of measure and unit of measure groups.
- In this illustration, we can see an example of how global units of measure can be grouped. A *Unit of measure group* is a set of units that are used for specific item types.
- Single units of measure can be grouped together as a subset. Then each Unit of Measure group has definitions of the relationships between the units inside the group based on conversion rules.
- The units of measure can be used in different groups and can be related to different items.
 - This means that each unit of measure may have a different meaning for different items. For example in this illustration, we see that a carton is used for multiple purposes. The Carton unit of measure is used in both the Paper group and the Media group.
 - Sometimes, when using the same unit for many items, it is better not to enter the volume and weight dimensions in the Units of Measure Setup window. We can enter the dimensions manually for each Item Master data.

Step 2: Define a Unit of Measure Group



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The unit of measure group for cables needs to contain all the units we use for purchasing, storing and selling the cables.

In step 1, we entered three unit of measure codes into the global units of measure list: meter, reel and spool.

Now we can group the three units together into a unit of measure group which we will ultimately assign to our cable items.

Since this unit of measure group will be used for cables, we call the group Cable.

The first unit entered will be the base unit. It is a best practice to use the unit used in inventory management as the base unit.

Then we define the relationship between the base unit and all other units in the group.

Unit of Measure Group windows

Setup → Inventory →
Units of Measure Groups

Unit of Measure Groups		
#	Group	Group Description
1	Manual	Manual
2	Paper	Paper
3	Cable	Cable

In the first row you choose the base unit.

Group Definition – Cable - Setup					
#	Alt Qty	Alt UoM	=	Base Qty	Base UoM
1	1	Meter	=	1	Meter
2	1	Reel	=	100	Meter
3	1	Spool	=	50	Meter

Then you enter alternative units of measure and their relationship to the base unit

These are the global units of measure we defined in the *Unit of Measure* setup window

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- Let's take a look at the Unit of Measure Group windows and settings
- In the *Units of Measure Groups* window, we create an entry for our cable group with a group code and description.
- Then choose the Group Definition button to open a window to define the relationship between units of measure within the group.
- Each group has a base unit. All the other units in the same group are related to the base unit by conversion rules.
- In our example, we enter the unit "Meter" as the base unit since this is the most commonly used unit for the cables. We track cable inventory by meters and we buy and sell all our cables in units of measure that are all based on the meter.
- Note that since the Meter is the base unit, on the first row the *Alternative UoM* is also set to Meter and the *Alternative quantity* and the *Base Quantity* columns are automatically set to "1" and are read only.
- Now we can add other units to our group, but they must always be related back to the base unit.
- In our example, we purchase the cables in reels. Therefore we enter Reel as an *Alternative UoM*.
- For each additional unit we add, we need to enter the fields alternative quantity and base quantity. One reel is equal to 100 meters of cable, so we set the alternative quantity to 1 and the base quantity to 100.

Step 3 Option 1: Assign UoM Group to Item Group

You can assign a unit of measure group as the default for an item group.

Item Group – Setup	
Item Group Name	Cables
General	
Default UoM Group	Cable
Default Inventory UoM	Meter

Once you assign a default UOM group, you can set a default for the Inventory unit of measure.

Be aware that this value cannot be changed in an item once there are existing transactions for that item.

- An optional step 3 is to define default values for the UoM group and Inventory UoM in your item groups.
- If you enter only the default UoM Group and not the inventory UoM, when adding a new item with this group the inventory UoM will be the base unit in the UoM group definition.
- In our business example, the purchasing coordinator can enter the Cable UoM group as the default for the Cables item group and specify a default inventory UoM.
- When a new item is added to the Cables item group, these default values are updated automatically in the item.
- When updating the default inventory unit of measure, a system message appears, asking if we want to update all existing items in this group as well. This update will take place only for items with no inventory transactions.
- Remember the Inventory unit of measure is the unit used in all inventory postings and it cannot be changed for an item once there are existing transactions for an item. This means that in some cases you will not want to set a default Inventory unit of measure but instead will want to define it manually for each item.

Step 3 Option 2: Assign UoM Group Directly to an Item

Paper UoM Group

- Meter = Base unit
- Reel = 100 Meters
- Spool = 50 Meters

Item Master Data

Item No.	Manual	C1001	☒ Inventory Item	
Description			☒ Sales Item	
Foreign Name			☒ Purchase Item	
Item Type	Items			
Item Group	Items			
UoM Group	Paper			
Price List	Base Price		Bar Code	Pack
			Unit Price	Primary Curre
			Pricing Unit	Pack

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- If your item groups include many items that do not share common units of measure, then it may make more sense to assign unit of measure groups directly to each newly created item.
- You can assign a unit of measure group manually to an item by choosing from the dropdown box in the *UoM Group* field in the *Item Master Data* window.

Step 4: Set Default Values in Item Master Data

Assign Inventory Unit on the **Inventory Data** tab

Set Inv. Method By	Warehouse
UoM Code	Meter
UoM Name	Meter
Weight	
Inventory Counting UoM Code	Meter
Inventory Counting UoM Name	Meter
Items per Counting Unit	1 Meter

You can also optionally assign an inventory counting unit on the Inventory Data tab

Assign a default Sales Unit on the **Sales Data** tab

Sales UoM Code	Meter	...
Sales UoM Name	Meter	
Items per Sales Unit	1	Meter

Assign a default Purchasing Unit on the **Purchasing Data** tab

Purchasing UoM Code	Reel	...
Purchasing UoM Name	Reel	
Items per Purchase Unit	100	Meter

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- Once an item master has an assigned unit of measure group, you make the following settings in the item master record:
 - Assign a inventory unit of measure – all inventory transactions will be made using this unit. This is setting is made in the *Inventory Data* tab. When you assign the unit of measure group to the item, the base unit of the group will default into this field automatically. You can change the inventory unit of measure assignment as long as there are no existing transactions for the item.
 - Also in the *Inventory Data* tab, you can optionally define a unit of measure for inventory counting.
 - Assign a default sales unit of measure on the Sales Data tab.
 - Assign a default purchasing unit of measure on the Purchasing Data tab.
 - Additionally, after a unit of measure group is assigned to an item master data, you can then optionally define an inventory counting unit of measure, barcodes, pricing and packaging for the item relating to each unit of measure within the group.

Recap: Menu paths for configuration of UoMs



Administration → *Setup* → *Inventory*

- Units of measure
- Units of measure group
- Length and Width
- Weight

Administration → *System Initialization* → *General Settings* → *Inventory* → *Items*

- Automatically add all UoM Group Definitions to New and Existing Items

There are several menu areas where you find unit of measure configuration settings.

Most of the configuration settings for unit of measure codes and groups is in the *Setup* menu in the *Administration* module. Here you can set up the unit of measure codes, the unit of measure groups and also characteristics that can be used to further define the codes such as length and width and weight.

In the *General Settings* menu under the *Inventory* tab, you will find settings for items that include the option to automatically add all Unit of Measure Group definitions to new and existing items.

Summary



Here are some key points:

- There are four main steps for setting up units of measures for items.
 1. Define the units of measure that will be used globally.
 2. Define Unit of Measure groups.
 3. Assign the unit of measure groups to item groups or directly to items.
 4. Set default values in the item for units of measure.
- All alternative units in the group are related to the base unit by conversion rules.
- Keep in mind that the inventory unit of measure for an item will be used for all inventory postings for that item. This value is not changeable for an item once transactions exist for that item.

There are four main steps for setting up units of measures for items.

- First, define the units of measure that will be used globally.
- Second, define units of measure groups. You do this by combining the units into unit of measure groups based on how items are managed when they are purchased, sold or managed in inventory.
- Assign the unit of measure groups to item groups or assign them directly to items. If you choose the latter, the unit of measure group will be assigned to an item automatically when it is created within an item group.
- Set additional default values in the item for units of measure for use in purchasing, sales, and inventory.

All the alternative units in the same unit of measure group are related to the base unit by conversion rules.

Keep in mind that the inventory unit of measure for an item will be used for all inventory postings for that item. This value is not changeable for an item once transactions exist for that item.

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